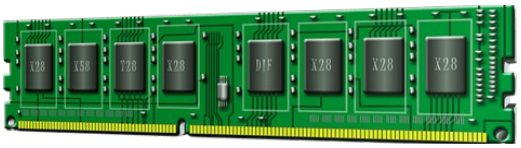
Memory

Introduction

Memory is where binary data (0s and 1s) is stored. There are several devices in a modern computer which store data. They store this data in different ways and for different purposes. We shall now take a look at each type of memory device:

- RAM
- ROM
- Virtual Memory

RAM Random Access Memory – AKA Main Memory

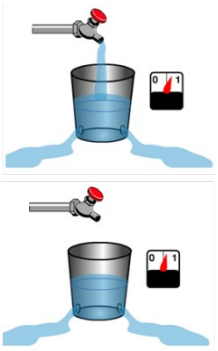


RAM is needed to store programs that are currently being used. They help to enable 'multi-tasking', which means having several programs open and using them all at the same time. It does this by copying the data needed by programs at that time so that it can be passed to the CPU when it needs to process that data. It also stores all the instructions/modules from the open programs that the CPU will require for processing.

RAM stores data as small charges of electricity in tiny transistors etched into a circuit in the device. The charge needs to be refreshed every few milliseconds otherwise the charge leaks away. RAM is therefore volatile memory – it loses data when there is no power.

When programs are opened by the user, they are loaded from the hard disk onto the RAM. This is so that the CPU can access the data and instructions at an acceptable rate. If the CPU reads the program instructions and data from a hard disk, the data access would be extremely slow – this is why RAM is so important.

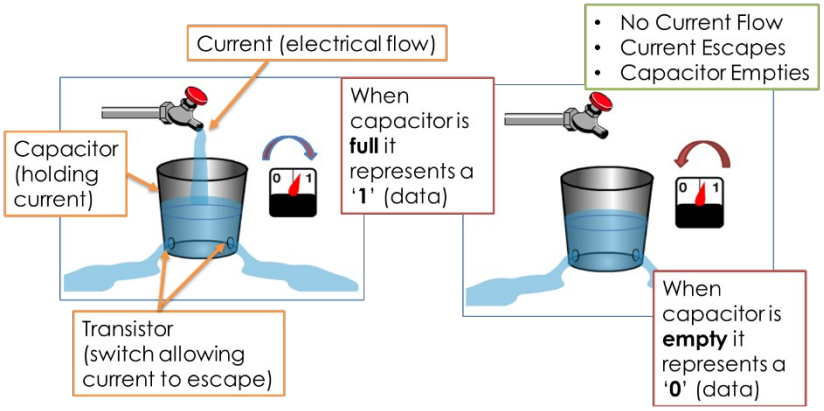
How does RAM store data?



Ram holds data using capacitors and transistors. A capacitor can be thought of as a sink or bucket holding water. When it is full it's holding a 'bit' of data (1), when it is empty it represents a zero (0). But obviously in RAM it holds an electrical current (not water). The transistor acts as a switch that lets the computer read the what is in the capacitor (bucket) or fill it up or empty it (change its state).

How Capacitors and Transistors Work – a brief overview!

Image Source: http://www.howstuffworks.com/ram.htm



Types of RAM

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Dynamic RAM

1 transistor and 1 capacitor for each bit – hold a small charge for a few milliseconds then needs to be refreshed. Because it needs refreshing, there is a limit as to how fast data can be read on this type of RAM. Power is always needed.

Static RAM

4-5 transistors wired together for each bit of data, but it does not lose its charge No need to refresh every few milliseconds – therefore much faster as they don't need refreshing. Does still need constant electricity supply. Faster but more expensive!

ROM - Read Only Memory



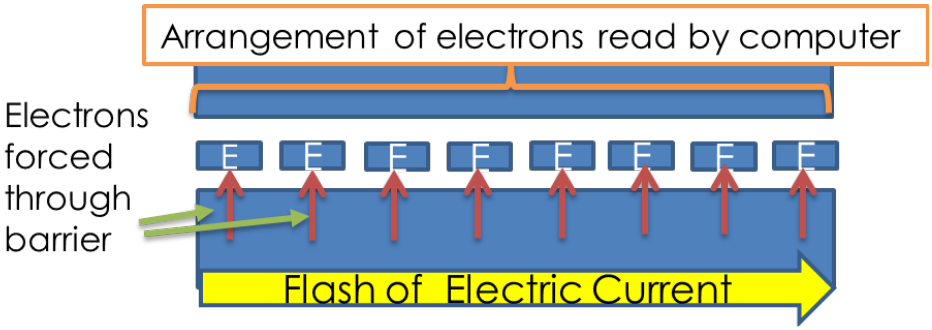
When you switch off your computer the data that the CPU can access (i.e. the RAM) loses its data. This is a problem because, when you restart your computer, data must be in the RAM for the computer to work. This is where ROM comes in.

ROM is a special memory chip that can store data even without power – it is non-volatile. It is used by a computer to store the boot sequence instructions – everything the computer needs to boot (start-up) and load the operating system. When you switch on your computer, data in the ROM is accessed by the CPU in order to start the computer and load the operating system successfully. The data on ROM cannot easily be over written by the user.

Like the RAM it is classed as Primary Storage (memory that is directly accessed by the CPU) & has fast data access rates. Modern ROM chips are now made of flash memory. This is a suitable type of storage technology because it doesn't need constant power supply to hold data.

How Flash Memory Works

Flash memory works by sending a large electric current to force electrons through a barrier and trapping them in position.



The flash of the electric current used to achieve this gives us the name for this type of memory. The different arrangement of electrons gives us data.

Virtual Memory



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If you are browsing the internet and listening to music, the RAM should cope fine. If you then decide to play a game with good graphics then this may overload the RAM as too many programs (and their data) will need to be loaded onto the RAM. So that the system doesn't grind to a halt and freeze up, the computer will relocate programs that have not been recently used to secondary storage (e.g.: the hard disk). This process will take time as hard-disk data speeds are much slower – but when it is done the system should be nice and fast again. Often the operating system will use the hard disk as virtual memory. A lot of programs need to be loaded as part of the operating system but many can sit on the hard-disk as they are used infrequently.

How the amount of RAM affects performance

So as we've just seen, RAM is needed to supply to the CPU with data and instructions from programs currently in use and when it gets full, the system runs slower as a result of the need to use virtual memory. Therefore the following is true:

- Smaller amounts of RAM – limited multi-tasking and greater need to use slower virtual memory
- Larger amounts of RAM – greater storage of programs / data leading to faster performance and effective multi-tasking.

Cache Memory

We have already seen cache memory when studying the workings of the CPU. We will now look at it in a little more detail.

As we have seen before, RAM will hold a copy of data and instructions (program) that are currently being used by the user. And unfortunately, as the CPU processes data faster than the RAM can supply it with data, system performance can suffer. This is where cache comes in...

Cache is simply a few Mb of memory that sits inside the CPU. Being located in the CPU means that data access speed is equal to the CPU's speed. CPU performance is therefore maintained. Well written programs will make sure that 'sets' of regularly used instructions are copied to the cache so that the program can run more efficiently and much faster. Poorly written programs will tend to need to fetch the instructions they require from the RAM one at a time and so, as access to the RAM is slow, the program performs less efficiently.

Questions *(The question zone you choose must either match your target grade or be higher!)*

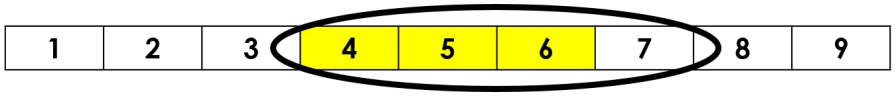
1	2	3	4	5	6	7	8	9

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Question Zone 1-3

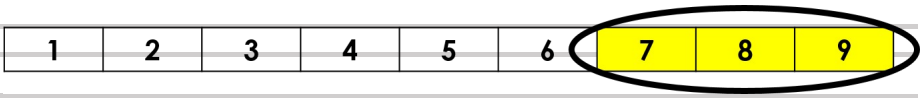
1. State the difference between 'volatile' and 'non-volatile' memory. [2]
2. Describe the purpose of RAM and ROM. [2]
3. What is virtual memory and why is it important. [3]

Question Zone 4-6



1. What is virtual memory and why is it important. [3]
2. Explain the effects of the amount of RAM a computer has on its performance. [4]
3. Explain, using a diagram, how flash memory technology works. [4]

Question Zone 7-9



1. Explain, using a diagram, how flash memory technology works. [4]
2. Explain, using a diagram, how RAM memory technology works. [5]
3. Discuss the relative importance of ROM, RAM and Virtual Memory in a computer system. [5]

Stick answer sheet here

State/Identify/Give/Name: Simply label a diagram, fill out a table or write a few words
Describe: Describing is 'saying what you see' (E.G.: A computer will have a CPU, Primary and Secondary storage etc)
Explain: Explaining is 'saying WHY/HOW something is like that'. (E.G.: A computer will have a CPU so that it can process all of the data the computer needs to perform a range of tasks. Primary and Secondary storage is needed because...)
Discuss: Discussing is 'looking at two sides of an issue, weighing up the two views and giving a conclusion'. Often these require a mini essay answer. (E.G.: New technology could be seen as being bad for the environment because..., but on the other hand, new technology has led to... In conclusion I believe that...)
Describe/Explain/Discuss **using examples**: Finally, if you are asked to give examples in any of these types of questions – YOU MUST **GIVE EXAMPLES**.

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Checklist:

- Date and title, clearly presented
- Spelling & grammar checked
- Question numbers in the margin
- Handwriting neat & legible
- Punctuation / Capital letters

Keywords / Key Terms:

RAM: *Random Access Memory – It is where programs are loaded when the user opens them. It is a volatile memory device.*

ROM: Read Only Memory – It is *a non-volatile memory device, which stores the boot up scripts.*

Virtual Memory: *When the RAM is full a proportion of secondary storage (often the hard disk) is used as an overflow allowing programs to continue to function, albeit slower.*

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Stick answer sheet here

State/Identify/Give/Name: Simply label a diagram, fill out a table or write a few words.
Describe: Describing is 'saying what you see' (E.G.: A computer will have a CPU, Primary and Secondary storage etc)
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Reflections **Score:** / **Percentage:** % **Grade:** **Progress** On / Above / Below

What Went

- ☐ I demonstrated a good level of understanding.
- ☐ I responded to the command words effectively.
- ☐ My answers were detailed / were written in depth.
- ☐ My work was well presented / legible.
- ☐ My answers effectively incorporated technical terminology.
- ☐ My responses were well structured / organised.
- ☐ My revision strategy was effective as I showed depth of understanding in my answers.
- ☐ My answers contained enough points / examples / explanations to achieve the marks available.

Even Better

- ☐ My answers need to be more accurate.
- ☐ I must respond correctly to the command words.
- ☐ My answers need more detail / greater depth.
- ☐ I must take greater care over my work / write neatly.
- ☐ I must incorporate key terminology into my answers.
- ☐ I must better organise my answers to improve its clarity.
- ☐ I need to improve my revision strategy as I did not demonstrate a depth of understanding in my answers.
- ☐ My answers didn't contain enough points / examples / explanations to achieve the marks available.

Further thoughts:

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